

Baggu Leelaprasad

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TECHNICAL SKILLS

- **Languages:** Python(Basic), C, HTML/CSS, VLSI
- **Frameworks:** MATLAB, Verilog, Arduino IoT, ESP-IDF
- **Databases:** SQL,
- **Tools:** GitHub, VS Code, ModelSim, VHDL, MATLAB, Arduino IDE
- **Others:** Network Analysis, Digital Electronics, Signals System

EXPERIENCE

VLSI & Digital Engineering Intern — SkillDzire Technologies Private Limited - May 2025 – June 2025

- Technologies:** Verilog HDL, ModelSim, EDA Playground, Digital VLSI, FPGA Design, FSM Modeling, Testbench Verification
- Completed comprehensive practical training in Digital VLSI Design methodologies and RTL modeling using Verilog HDL.
 - Developed synthesizable digital modules including traffic priority controllers, synchronous FIFO buffers, and finite state machines(FSM).
 - Conducted functional verification and created self-checking testbenches in ModelSim to validate setup/hold margins and corner cases.
 - Explored FPGA architecture fundamentals including Look-Up Tables (LUTs), Configurable Logic Blocks (CLBs), and Vivado synthesis flows.
 - Earned official Certificate of Completion (Credential ID: SDST-25-13055).

Data Analysis Using Python Intern — APSSDC - Summer 2025

- Technologies:** Python 3.x, NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebook, Exploratory Data Analysis
- Completed intensive industrial training in Data Analysis Using Python organized by Andhra Pradesh State Skill Development Corporation (APSSDC).
 - Engineered data analytics pipelines using NumPy, Pandas, Matplotlib, and Seaborn for multi-dimensional data exploration.
 - Analyzed historical 120-year Olympics dataset to extract medal trends, nation-wise performance curves, and sport dominance distributions.
 - Presented analytical findings through publication-grade interactive graphs, statistical heatmaps, and Jupyter documentation.

PROJECTS

Smart Traffic System for Emergency Vehicles Using IoT – Verilog HDL, FPGA, RFID, IR Sensors, FSM, IoT

- Developed an automated traffic preemption controller using **Verilog HDL** and FPGA-based FSM architecture to provide priority to emergency vehicles.
- Integrated **RFID and IR sensors** to detect emergency vehicles and estimate real-time traffic density across intersection lanes.
- Designed a deterministic priority controller with safe yellow transition, all-red clearance, emergency green corridor, and automatic recovery to normal traffic scheduling.
- Simulated and verified the controller using **ModelSim and EDA Playground** with FPGA synthesis using Vivado/Quartus.

Olympics Historical Data Analysis & Visual Analytics – Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

- Developed a comprehensive data analysis pipeline using **Python, Pandas, NumPy, Matplotlib, and Seaborn** to analyze historical Olympic datasets.
- Cleaned and processed athlete-level data to identify **medal trends, country-wise performance, sport dominance, and participation patterns**.
- Generated country-wise cumulative medal progression graphs, statistical visualizations, correlation matrices, and time-series analysis.
- Documented analytical findings and visual insights using **Jupyter Notebook**.

EDUCATION

B.Tech in Electronics and Communication — GIET Engineering College, Rajahmundry (2023–2027) - **CGPA: 7.9/10**

EXTRACURRICULAR ACTIVITIES

- NSS Volunteer at GIET Engineering College
- Runner-up in Inter-College Volleyball Competition

CERTIFICATIONS

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| • Certified in IOT(NPTEL) — Elite, 70% | –Jan-Apr 2026 | • Certified in AIML | –May-July 2026 |
| • Certified in VLSI(SkillDzire) | –May-Jun 2025 | • Certified in Data Analysis Using Python | –May-july 2025 |